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Short Communication

First report of a multidrug-resistant *Salmonella enterica* Serovar Infantis carrying pESI megaplasmid isolated from marine shrimp in India

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ABSTRACT

Background:: Non-typhoidal *Salmonella enterica* (NTS) in seafood is an important human health concern. An emerging strain of NTS serovar Infantis carrying a megaplasmid pESI and resistant to multipe drugs has been responsible for frequent food-borne human infections worldwide.
Methods:: *S. enterica* strain JS5 was isolated from a sample of shrimp from the retail market on XLD agar after enrichment in the RV medium. The genomic DNA was isolated and sequenced using the Illumina platform. The draft whole genome sequence of *Salmonella* Infantis JS5 revealed the presence of a plasmid.
Results:: The genome size was 4,977,731 bp with 4663 open reading frames. The bacterium harboured a megaplasmid similar to the pESI plasmid reported in the emerging *S.* Infantis. The 285 kb plasmid contained the characteristic genes of the pESI plasmids, such as the mercury operon, yersiniabactin siderophore operon, fimbriae, toxin-antitoxin systems and the hypothetical protein backbone. The antibiotic resistance genes *tet*(A), *dfrA14, aadA, qacEdelta1*, and *sul1* were detected. To the best of our knowledge, this is the first report on pESI plasmid carrying *S.* Infantis from India.

1. Introduction

Non-typhoidal *Salmonella enterica* (NTS) is one of the leading causes of foodborne diarrhoeal diseases worldwide. Over 1500 serovars of NTS are commonly associated with diverse animals and birds and can easily contaminate foods originating from these sources. *Salmonella enterica* is not a common inhabitant of coastal-marine waters, and its presence in seafood is attributed to the contamination of seafood harvesting areas with human and animal wastes. Between 1998 and 2005, fish was the most commonly implicated food category in outbreaks, with *Salmonella* causing most hospitalisations (31%) involving large outbreaks, with the victim numbers ranging from 3 to 425 in the USA [1]. Seafood contamination with NTS has been frequently reported in India [2,3], with *S.* Weltevreden, *S.* Rissen, *S.* Typhimurium and *S.* Derby being some of the most predominant NTS serovars in seafood [4].

Salmonella enterica subsp. *enterica* serovar Infantis, commonly associated with poultry, has been gaining attention as an emerging pathogen involved in frequent foodborne outbreaks. The presence of pESI (Plasmid Emergent *Salmonella Infantis*) conjugative megaplasmid confers resistance to multiple antimicrobials and virulence without burdening the host bacterium [5]. Following its first report from Israel, the pESI plasmid was subsequently reported in *S.* Infantis from several countries as one of the fast-spreading resistance plasmids [6]. This study reports the draft genome sequence of *S.* Infantis harbouring pESI-like conjugative plasmid. To our knowledge, this is the first report of a multidrug-resistant *S.* Infantis with a pESI-like plasmid from India.

2. Materials and methods

2.1. Bacterial isolation and whole-genome sequencing

Salmonella was isolated from a marine shrimp sample collected from a local market in Mumbai, India, in 2019 using the standard protocol of pre-enrichment in lactose broth, followed by selective enrichment in Rappaport Valisladis medium and Tetrathionate broth at 42±0.5 °C [7]. For isolation, Xylose lysine

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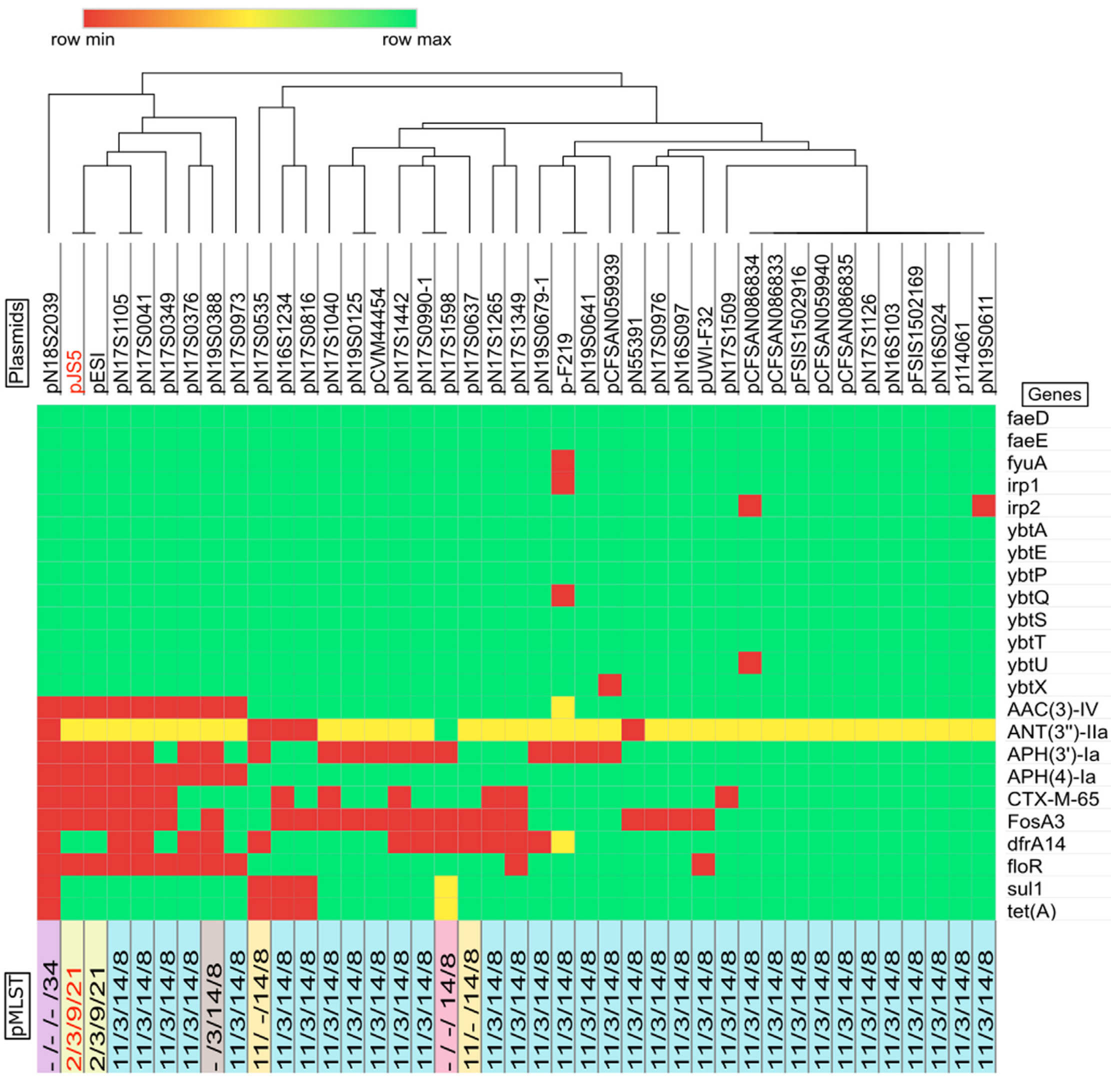

Fig. 1. pESI plasmids (n = 40) retrieved from the NCBI were clustered together using the Euclidian distance approach. The gene names are on the upper; the lower x-axis consists of the pMLST of the respective plasmid in the gene order *ardA, pill, sogS, trbA*, respectively. The green and red boxes indicate the presence and absence of genes corresponding to the genes indicated on the y-axis and the plasmids on the upper x-axis. The yellow coloured boxes signify query coverage of less than 100%. The heatmap was generated using Morpheus (https://software.broadinstitute.org/morpheus).

deoxycholate agar, Hektone enteric agar and Bismuth sulphite agar (Himedia, Mumbai, India) were used. The isolate was confirmed as *Salmonella* by PCR for *invA* and *hilA* genes [8]. The DNA of JS5 was isolated from the overnight broth culture using the CTAB phenol-chloroform method [9]. For the whole-genome sequencing, library preparation was done using TruSeq Nano DNA Kit (TruSeq Nano DNA Sample Preparation Guide, Part # 15,041,110 Rev. D) and sequenced using Illumina platform generating paired end reads. SPAdes assembler version 3.14.1 was used to assemble the draft genome with default parameters [10], and the assembly polishing was performed using the MeDuSa web tool [11]. The assembled genome was annotated using the prokaryotic genome annotation tool Prokka [12] and RAST[13].

2.2. Plasmid assembly and annotation

MOB-suit standalone tool [14] was used to predict the plasmid sequences from the de-novo assembled contigs. PlasmidFinder [15] and pMLST web tools [14] were used to identify plasmid replicon and plasmid typing. The RAST [13] and PATRIC [17,18] servers were used to annotate and explore the pESI plasmid of this study designated as pJS5. The IncFIB replicon was used as a query in the PLSDB database [16], and 40 pESI plasmid sequences of different *Salmonella* Infantis were retrieved from NCBI. All the plasmid assemblies were screened for antimicrobial resistance genes (CARD) [19] as well as the virulence gene (VFDB) [20] using Abricate (https://github.com/tseemann/ABRicate). The results were used to generate a heatmap. The mobile genetic elements were analysed using VRprofile2 [21] and Integron Finder [22].

3. Results and discussion

3.1. Bacterial isolation and whole-genome assembly

Salmonella Infantis strain JS5 was isolated on XLD agar after enrichment in the RV medium. The final draft assembled genome consisted of 19 scaffold sequences, with an N50 value of 4,770,289 bp and a GC content of 52.15%. Of 4760 genes predicted, 4663 were protein-coding genes comprising 3897 characterised proteins and 766 putative/hypothetical proteins. Of 96 protein non-coding genes, 14 were rRNA genes, and 82 were tRNA genes. Further, 3196 proteins could be annotated to various pathways.

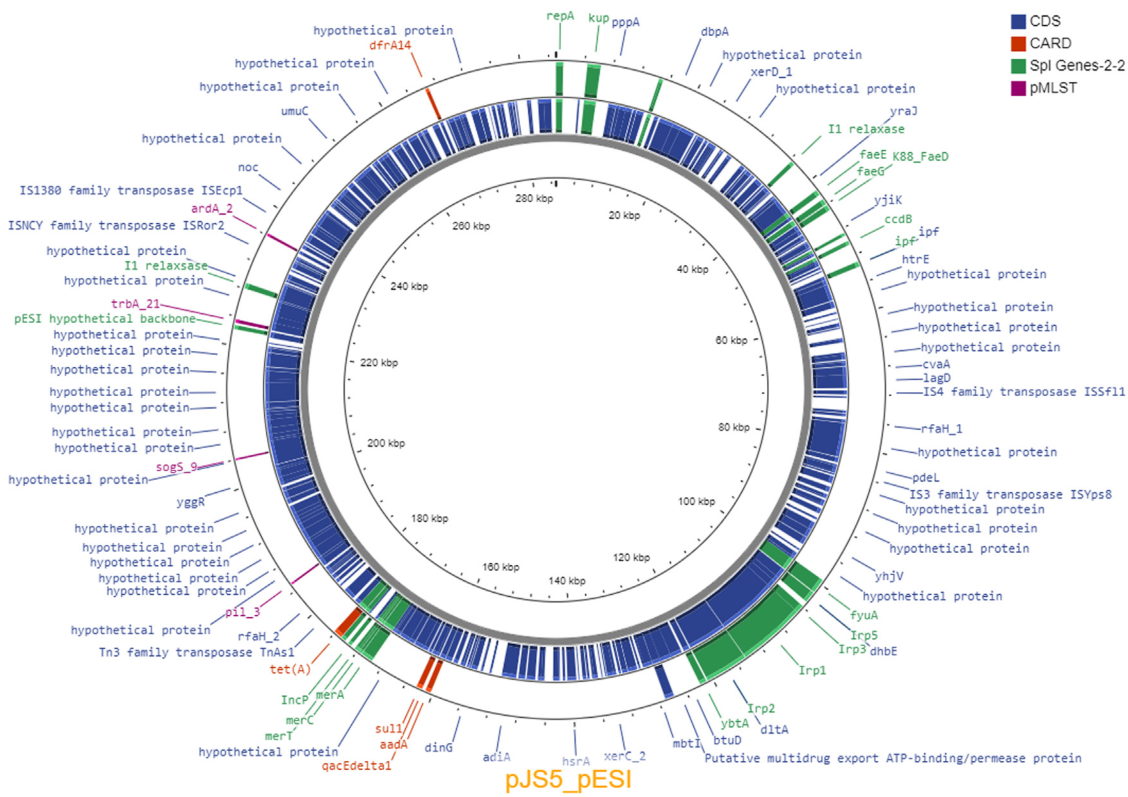

Fig. 2. The plasmid pJS5 is visualised using the Proksee server (https://beta.proksee.ca/). The pESI plasmid related gene sequences were retrieved from NCBI and blasted against the pJS5. The inner circle is the ~285 kb long plasmid contig. The next track shows the CDC annotated by prokka in blue. The outermost track shows the genes of interest. The red region contains antibiotic resistance genes found by CARD; the purple regions are the pMLST genes detected; the green regions contain genes typical to the pESI plasmids.

3.2. Plasmid assembly and annotation

The plasmid contigs were separated using MOB-suit [14], resulting in 9 contigs with an N50 of 62,754 bases. The plasmid size was estimated to be 285 kb. The RAST server annotation revealed 413 coding sequences and 7 subsystems. The pJS5 replicon was of IncFIB9 (pN55391), characteristic of the pESI megaplasmid.

The pMLST profile of pJS5 was found to be Incl1, the nearest being 209 ST. The four pMLST genes analysed, namely *ardA*, *pill*, *sogS*, *trbA*, gave a pMLST allele type of 2/3/9/21. This pMLST type was found in the first pESI plasmid reported [6] (Accession: NZ_CP047882.1). A study [23] comparing and grouping the pESI plasmids based on pMLST found specific genetic characteristics conserved in the particular group. The pESI plasmids with pMLST 2/3/9/21 allele type generally lack the $bla_{CTX-M65}$, *floR*, *aac(3)-IV*, *aph(4)-la*, *fosA3* genes [23], which was also observed in pJS5.

The plasmid pJS5 was clustered with 40 *Salmonella* Infantis pESI plasmid sequences retrieved from the GenBank and compared for the antimicrobial resistance and virulence genes (Fig. 1). The pJS5 and the pESI reference plasmid clustered together. There was no significant difference between the metabolic models reconstructed from pJS5 and the pESI plasmid sequences (Accession: NZ_CP047882.1) using the SEED viewer [6]. The blastn comparison showed 99.21% identity between pJS5 and pESI plasmids with 99% query coverage.

The genes characteristic of pESI plasmids were found on the JS5 plasmid, which included a mercury resistance operon (*merA*, *merC*, *merE*, *merT*), siderophore Yersinibactin operon (*Irp1*, *Irp2*, *Irp3*, *Irp4*, *ybtA*, *fuyA*), three toxin-antitoxin systems namely the MazEF system containing the genes *pemK*, *pemI*, the *ccdA*, *ccdB* system and the *vapB*, *vapC* system. The *dinJ* antitoxin that binds with the toxin *yafQ* was also present. Apart from these, the tellurite resistance gene *tehA*, and potassium homeostasis *kup* gene were also found. The annotated and manually curated plasmid pJS5 is shown in Fig. 2. The hypothetical backbone protein reported in very few pESI plasmids [24] was found in pJS5.

The antimicrobial resistant genes *tet*(A), *dfrA14*, *aadA*, *qacEdelta1*, *sul1* genes were found on the JS5 plasmid. The genotypic resistant profile of this plasmid matched with the phenotypic resistance of the *S.* Infantis JS5. The phenotypic resistance to tetracycline could be matched with the presence of the *tet(A)* gene, resistance to neomycin (aminoglycoside) to that of *aadA* gene and resistance to sulfamethizole and sulphaphenazole to the *sul1* gene. Ethidium bromide at 10 mg/mL did not affect the survival of JS5 (data not shown), which could be attributed to the *qacEdelta1* gene (data not shown). These antimicrobial genes are known to be associated with mobile genetic elements. A complete class 1 integron (9521 bp) and an incomplete integron was found on the plasmid pJS5. Four Insertion sequences and two composite transposons IS6 and IS3 were involved in the antimicrobial gene - mobile genetic elemet clusters. The *aadA*, *qacEdelta1*, *sul1* genes were associated with the ISSen7/IS26/IS1326 Transposase. The *tet(A)* gene was associated with the TnpA_TnAs1 and *dfrA14* gene with the IS26 Transposase. A comprehensive study on pESI plasmids [24] showed that the bla_{CTX-M} gene was found in only 49% of the 654 pESI plasmids data analysed. Though the bla_{CTX-M} gene was absent in pJS5, phenotypic resistance to 3rd generation cephalosporins, penem and penams was observed, which could be attributed to many efflux pump genes annotated on the chromosomal assembly (data not shown).

4. Conclusion

The occurrence of pESI carrying *S*. Infantis in India is of concern and suggests domestic sources of contamination such as poultry [24], although the occurrence of pES1-carrying *S*. Infantis in poultry has not been reported so far from India. The widely consumed seafood can act as a vehicle for community transmission of antibiotic-resistant *S.* Infantis. More surveillance studies are necessary to understand the dissemination of pESI carrying *S*. Infantis in the Indian subcontinent.

Data availability

The whole-genome draft assembly is submitted in NCBI under BioProject accession number PRJNA794263. The whole-genome assembly is submitted to the genebank under accession number *Salmonella enterica* subsp. *enterica* serovar Infantis JS5 (JAKCLH000000000; BioSample SAMN24619174).

Ethical approval

Not required.

Competing interests

None declared.

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